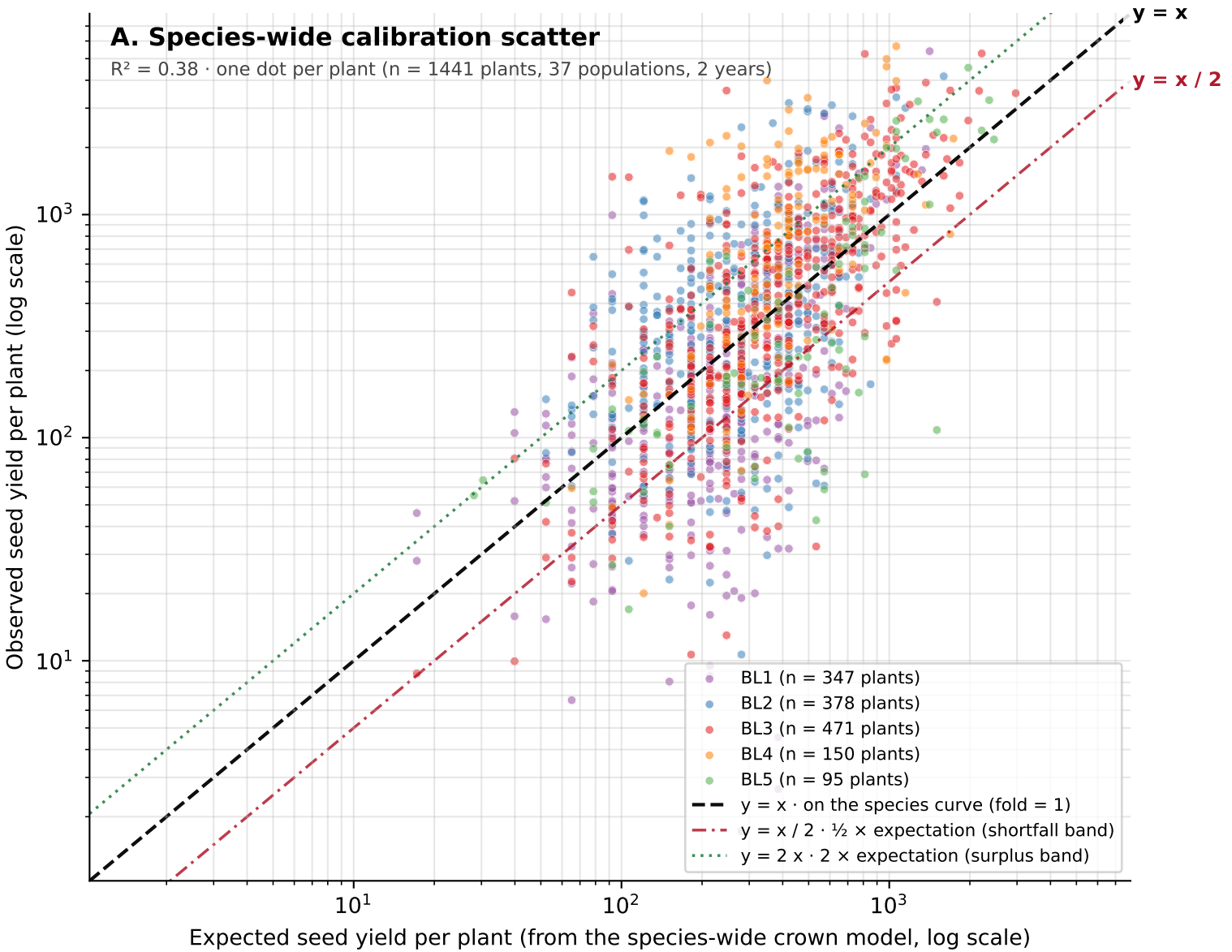
Species-wide plant-size → seed-yield expectation (the yardstick for the per-population seed-yield test)



B. Model build — how the species-wide expectation is set

STAGE 1 · predictor selection (10-fold cross-validation)

Lower CV-RMSE = better out-of-sample fit → wins Stage 2.

predictor	CV-RMSE	±sd	R²_in	AIC
crown	0.411	0.035	0.384	1540.3 ⋆
crown+height	0.412	0.035	0.384	1542.2
area	0.421	0.031	0.356	1603.5
height	0.451	0.027	0.261	1801.0

→ Winner: crown (★)

STAGE 2 · refitted expectation model (all plants pooled)

Fitted value = expected $\log_{10}$(yield) for a plant of that size.
Residual = observed − expected.

Formula : $\log_{10}$(yield) ~ logc
Coefficients : Intercept = +1.236, logc = +1.212
R^2 (in sample): 0.384
Residual SD : 0.413 $\log_{10}$ (≈ × / ÷ 2.6)

Every population in the per-population test is measured against this expectation.